

Laboratory Work 4

DQL

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The screenshot displays a PostgreSQL IDE interface. On the left, the 'Object Explorer' shows a tree view of the database structure, including 'lab4', 'public', and 'airline' schemas. The main window shows a SQL query with 15 tasks, each preceded by a comment line (e.g., '--task 1'). The query includes various DQL operations such as SELECT, WHERE, JOIN, and ORDER BY. The status bar at the bottom indicates 'Total rows: 50' and 'Query complete 00:00:00.063'.

```
1  --lab4
2  --task 1
3  SELECT UPPER(airline_name) AS upper_airline_name FROM airline;
4  --task 2
5  SELECT REPLACE(airline_name, 'Air', 'Aero') FROM airline;
6  --task 3
7  SELECT flight_no FROM flights WHERE airline_id=1
8  INTERSECT
9  SELECT flight_no FROM flights WHERE airline_id=2;
10 --task 4
11 SELECT airport_name FROM airport WHERE airport_name LIKE '%Reginal%' AND airport_name LIKE '%Air%';
12 --task 5
13 SELECT first_name, last_name, TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS formatted_birth_date FROM passengers;
14 --task 6
15 SELECT flight_no FROM flights WHERE actual_arrival > scheduled_arrival;
16 --task 7
17 SELECT * FROM flights WHERE actual_arrival > scheduled_arrival;
18 --task 8;
19 SELECT * FROM airline WHERE airline_country IN('France', 'Portugal', 'Poland')
20 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
21 --task 9
22 SELECT * FROM baggage WHERE weight_in_kg > 25 ORDER BY weight_in_kg DESC LIMIT 3;
23 --TASK 10
24 SELECT first_name || ' ' || last_name AS full_name FROM passengers ORDER BY date_of_birth DESC LIMIT 1;
25 -- task 11
26 SELECT booking_platform, MIN(price) AS cheapest_price FROM booking GROUP BY booking_platform;
27 --TASK 12
28 SELECT * FROM airline WHERE airline_code ~ '[0-9]';
29 -- task 13
30 SELECT * FROM airline ORDER BY created_at DESC LIMIT 5;
31 --TASK 14
32 SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300
33 AND check_result <> 'Checked';
34 --task 15
35 SELECT * FROM baggage_check WHERE EXTRACT(YEAR FROM update_at) = EXTRACT(YEAR FROM created_at)
36 AND EXTRACT(MONTH FROM update_at) = EXTRACT(MONTH FROM created_at)
37 AND update_at < created_at;
38
39
```

Total rows: 50 Query complete 00:00:00.063 LF Ln 3, Col 63

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post... X

lab4/postgres@PostgreSQL 18

Data Output Messages Notifications

Showing rows: 1 to 50 Page No: 1 of 1

	upper_airline_name
1	IPC
2	PDN
3	KLE
4	KHS
5	YLQ
6	NGL
7	0
8	QIG
9	NQX
10	SOZ
11	IVA
12	KQO
13	IFH
14	PIR
15	KKG
16	DUC
17	YLP
18	KIQ
19	CSC
20	SSL
21	YDQ
22	HGZ
23	SPI
24	RUG
25	CLY
26	YHB
27	KKL

Total rows: 50 Query complete 00:00:00.034 LF Ln 3, Col 63

TASK 2

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post... X

lab4/postgres@PostgreSQL 18

Query Query History

```
--task 2
SELECT REPLACE(airline_name,'Air','Aero') FROM airline;
--task 3
```

Data Output Messages Notifications

Showing rows: 1 to 50 Page No: 1 of 1

	replace
1	IPC
2	PDN
3	KLE
4	KHS
5	YLQ
6	NGL
7	0
8	QIG
9	NQX
10	SOZ
11	IVA
12	KQO
13	IFH
14	PIR
15	KKG
16	DUC
17	YLP
18	KIQ
19	CSC
20	SSL
21	YDQ
22	HGZ
23	SPI
24	RUG

Total rows: 50 Query complete 00:00:00.039 LF Ln 8, Col 1

TASK3

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, including the 'lab4' schema and the 'airline' table. The main query editor shows the following SQL code:

```
--task 3
SELECT flight_no FROM flights WHERE airline_id=1
INTERSECT
SELECT flight_no FROM flights WHERE airline_id=2;
--task 4
```

The Data Output pane shows the result of the query:

flight_no
character varying (50)

The status bar at the bottom indicates "Total rows: 0" and "Query complete 00:00:00.049".

TASK 4

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, including the 'lab4' schema and the 'airline' table. The main query editor shows the following SQL code:

```
--task 4
SELECT airport_name FROM airport WHERE airport_name LIKE '%Reginal%' AND airport_name LIKE '%Air%';
--task 5
SELECT first_name,last_name,TO_CHAR(date_of_birth,'FMMonth DD, YYYY') AS formatted_birth_date FROM passengers;
--task 6
```

The Data Output pane shows the result of the query:

airport_name
character varying (50)

The status bar at the bottom indicates "Total rows: 0" and "Query complete 00:00:00.060".

TASK 5

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18 X L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query Query History

```
--task 4
11 SELECT airport_name FROM airport WHERE airport_name LIKE '%Reginal%' AND airport_name LIKE '%Air%';
12 --task 5
13 SELECT first_name,last_name,TO_CHAR(date_of_birth,'FMMonth DD, YYYY') AS formatted_birth_date FROM passengers;
14 --task 6
```

Data Output Messages Notifications

Showing rows: 1 to 200 Page No: 1 of 1

	first_name character varying (50)	last_name character varying (50)	formatted_birth_date text
1	Hilde	Irnis	January 03, 2000
2	Arvy	Sparsholt	June 09, 1974
3	Reinald	Pococke	June 07, 1982
4	Con	Borrei	October 17, 1986
5	Wayne	Bangs	April 22, 1996
6	Tildy	Shackleford	April 15, 2004
7	Byrle	Oram	July 07, 1985
8	Howard	Igo	March 29, 1984
9	Alphonso	Philippou	August 30, 1975
10	Fairfax	Needham	July 08, 1981
11	Vanda	Prime	May 17, 1977
12	Terra	Wimmer	January 02, 1977
13	Domenic	Blaszczak	December 31, 2001
14	Conrad	Southerell	January 25, 1998
15	Heloise	Yukhin	June 09, 1995
16	Artur	Leap	January 19, 2008
17	Wood	Yve	January 18, 1997
18	Antonella	Lambdon	February 28, 1972
19	Muhammad	Fass	May 31, 1979
20	Melvyn	Daskiewicz	December 31, 1970
21	Marillin	Jewess	May 19, 1999
22	Aviva	Mauro	January 16, 2004

Successfully run. Total query runtime: 39 msec. 200 rows affected. X

Total rows: 200 Query complete 00:00:00.039 LF Ln 13, Col 111

TASK 6

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18 X L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query Query History

```
--task 6
15 SELECT flight_no FROM flights WHERE actual_arrival > scheduled_arrival;
16 --task 7
17 SELECT * FROM flights WHERE actual_arrival > scheduled_arrival;
18 --task 8;
```

Data Output Messages Notifications

Showing rows: 1 to 509 Page No: 1 of 1

	flightNo character varying (50)
1	US-CT
2	US-NM
3	RU-KR
4	US-AZ
5	IN-OR
6	CA-NL
7	BR-PE
8	TH-32
9	US-MS
10	GB-ENG
11	UZ-SU
12	US-OR
13	KP-10
14	MZ-G
15	US-AK
16	AU-NT
17	PG-MPL
18	TW-TTT
19	ID-BA
20	US-KS
21	CM-OU
22	ZA-FS

Successfully run. Total query runtime: 50 msec. 509 rows affected. X

Total rows: 509 Query complete 00:00:00.050 LF Ln 15, Col 72

TASK 7

Object Explorer

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post... X

lab4/postgres@PostgreSQL 18

Query Query History

```
--task 6
15 SELECT flight_no FROM flights WHERE actual_arrival > scheduled_arrival;
--task 7
17 SELECT * FROM flights WHERE actual_arrival > scheduled_arrival;
--task 8;
```

Data Output Messages Notifications

Showing rows: 1 to 509 Page No: 1 of 1

flight_id [PK] integer	flight_no character varying (50)	scheduled_departure date	scheduled_arrival date	departure_airport_id integer	arrival_airport_id integer	departing_gate character varying (50)	arriving_gate character varying (50)	air int
1	1 US-CT	2024-01-22	2023-09-08	12	15	9	649	
2	2 US-NM	2023-07-21	2023-09-17	13	16	4216	90	
3	4 RU-KR	2024-01-02	2023-03-18	3	19	151	66	
4	8 US-AZ	2023-07-29	2023-04-08	13	6	5045	1985	
5	9 IN-OR	2023-05-18	2023-09-19	6	9	659	621	
6	12 CA-NL	2023-03-22	2023-06-04	12	13	40	2592	
7	13 BR-PE	2024-01-16	2023-06-02	13	20	1891	5429	
8	18 TH-S2	2024-02-05	2023-03-30	8	19	478	[null]	
9	20 US-MS	2024-02-06	2024-01-22	10	1	346	56	
10	22 GB-ENG	2023-11-03	2023-10-11	2	2	400	85	
11	27 UZ-SU	2023-05-13	2023-05-24	11	2	1027	190	
12	30 US-OR	2024-03-10	2023-07-14	6	14	214	285	
13	32 KP-10	2024-03-12	2023-06-06	2	2	4547	2999	
14	33 MZ-G	2023-09-21	2023-11-29	4	19	[null]	9380	
15	34 US-AK	2024-02-08	2023-10-20	11	3	38	13	
16	36 AU-NT	2023-03-29	2023-05-06	12	14	[null]	13	
17	39 PG-MPL	2023-11-23	2023-06-05	12	12	2950	505	
18	40 TW-TTT	2023-07-04	2023-06-02	1	17	28	29	
19	41 ID-BA	2023-12-30	2023-04-04	2	9	14	5586	
20	42 US-KS	2023-03-25	2023-09-01	17	6	1691	973	
21	43 CM-OU	2023-12-01	2023-03-21	10	16	3963	10	
22	46 ZA-FS	2023-05-08	2023-06-05					

Successfully run. Total query runtime: 57 msec. 509 rows affected.

Total rows: 509 Query complete 00:00:00.057

TASK 8

Object Explorer

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post... X

lab4/postgres@PostgreSQL 18

Query Query History

```
17 SELECT * FROM flights WHERE actual_arrival > scheduled_arrival;
18 --task 8;
19 SELECT * FROM airline WHERE airline_country IN('France','Portugal','Poland')
20 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';
21 --task 9
```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	4 OCKB	KHS	France	2023-11-18	2024-02-18
2	13 OIFE	IFH	Portugal	2024-01-09	2023-11-09
3	15 SYKZ	KKG	Poland	2023-12-24	2023-10-30
4	24 ZSRG	RUG	Portugal	2024-03-11	2023-08-24
5	28 RRPD	JDG	Poland	2023-11-03	2023-12-06
6	30 CYST	YST	Poland	2023-11-24	2023-10-07
7	43 AYKM	KMA	France	2023-11-21	2023-05-30

Successfully run. Total query runtime: 53 msec. 7 rows affected.

Total rows: 7 Query complete 00:00:00.053

TASK 9

Object Explorer

- Languages
- Publications
- Schemas
- Subscriptions
- lab4
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (10)
 - airline
 - Columns (6)
 - airline_id integ
 - airline_code of
 - airline_name c
 - airline_countr
 - created_at dat
 - update_at date

lab4/postgres@PostgreSQL 18

Query

```
--Task 9
21
22 SELECT * FROM baggage WHERE weight_in_kg>25 ORDER BY weight_in_kg DESC LIMIT 3;
23
24 --TASK 10
25 SELECT first_name || ' ' || last_name AS full_name FROM passengers ORDER BY date_of_birth DESC LIMIT 1;
26 --Task 11
```

Data Output

baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer
1	321	2023-07-07	2023-11-10	349
2	61	2023-04-21	2023-07-31	464
3	102	2024-01-24	2023-11-01	53

Showing rows: 1 to 3 Page No: 1 of 1

Successfully run. Total query runtime: 42 msec. 3 rows affected

Total rows: 3 Query complete 00:00:00.042

TASK 10

Object Explorer

- Languages
- Publications
- Schemas
- Subscriptions
- lab4
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (10)
 - airline
 - Columns (6)
 - airline_id integ
 - airline_code of
 - airline_name c
 - airline_countr
 - created_at dat
 - update_at date

lab4/postgres@PostgreSQL 18

Query

```
--task 9
21
22 SELECT * FROM baggage WHERE weight_in_kg>25 ORDER BY weight_in_kg DESC LIMIT 3;
23
24 --TASK 10
25 SELECT first_name || ' ' || last_name AS full_name FROM passengers ORDER BY date_of_birth DESC LIMIT 1;
26 --Task 11
```

Data Output

full_name text
Bernie Mic...

Showing rows: 1 to 1 Page No: 1 of 1

Successfully run. Total query runtime: 36 msec. 1 rows affected

Total rows: 1 Query complete 00:00:00.036

TASK 11

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query Query History

```

25 -- task 11
26 SELECT booking_platform ,MIN(price) AS cheapest_price FROM booking GROUP BY booking_platform;
27 --TASK 12
28 SELECT * FROM airline WHERE airline_code ~ '[0-9]';
29 -- task 13

```

Data Output Messages Notifications

Showing rows: 1 to 494 Page No: 1 of 1

	booking_platform character varying (50)	cheapest_price numeric
1	Schmitt-Quigley	4966.58
2	Rohan, Hamill and Bosco	7118.84
3	Kutch-Russel	2113.46
4	Veum-Corwin	2183.14
5	Wolff, Wuckert and Satterfield	4911.89
6	Reichel, Deckow and Quitzon	8389.34
7	Murazik-Gulgowski	950.80
8	Cole, Corwin and Leannon	8735.70
9	Franecki, Dickens and Walsh	283.78
10	McLaughlin, Sipes and Abernathy	7490.57
11	Leffler-Thompson	2559.77
12	Raynor Inc	410.47
13	Waelchi Inc	3694.97
14	Renner, Weber and Wilkinson	4230.67
15	Langworth-Osinski	4982.91
16	Will LLC	6462.07
17	Stroman, Block and Halvorson	938.92
18	Mraz, Daniel and Schmeler	2077.18
19	Wisozk and Sons	868.47
20	Wuckert, Weber and Walter	2520.04
21	Treutel, Zemlak and Jast	7379.88
22	Graham, Torp and Mante	7813.55

✓ Successfully run. Total query runtime: 46 msec. 494 rows affected. X

Total rows: 494 Query complete 00:00:00.046 LF Ln 26, Col 95

TASK 12

lab4/postgres@Po... X lab4/postgres@PostgreSQL 18* X L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query Query History

```

25 -- task 11
26 SELECT booking_platform ,MIN(price) AS cheapest_price FROM booking GROUP BY booking_platform;
27 --TASK 12
28 SELECT * FROM airline WHERE airline_code ~ '[0-9]';
29 -- task 13

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	22	2AK6	HGZ	Philippines	2024-03-13	2023-10-20

✓ Successfully run. Total query runtime: 53 msec. 1 rows affected. X

Total rows: 1 Query complete 00:00:00.053 LF Ln 28, Col 52

TASK 13

Object Explorer | lab4/postgres@Po... | lab4/postgres@PostgreSQL 18* | L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query | Query History

```

29 -- task 13
30 SELECT * FROM airline ORDER BY created_at DESC LIMIT 5;
31 --TASK 14
32 SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300
33 AND check_result <> 'checked';

```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	9	KNQX	NQX	Venezuela	2024-03-16	2023-09-16
2	40	LTBD	CII	Argentina	2024-03-15	2024-01-29
3	22	2AK6	HGZ	Philippines	2024-03-13	2023-10-20
4	24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
5	45	HCMV	BUO	Sweden	2024-02-18	2023-12-10

Successfully run. Total query runtime: 47 msec. 5 rows affected

Total rows: 5 | Query complete 00:00:00.047

TASK 14

Object Explorer | lab4/postgres@Po... | lab4/postgres@PostgreSQL 18* | L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query | Query History

```

30 SELECT * FROM airline ORDER BY created_at DESC LIMIT 5;
31 --TASK 14
32 SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300
33 AND check_result <> 'checked';

```

Data Output | Messages | Notifications

Showing rows: 1 to 56 | Page No: 1 of 1

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	7	Not checked	2023-05-04	2023-04-29	226	44
2	12	Not checked	2023-10-02	2023-06-22	288	133
3	22	Not checked	2023-06-14	2023-11-10	246	101
4	23	Not checked	2023-05-29	2023-06-02	284	22
5	24	Not checked	2024-02-17	2023-10-03	267	68
6	25	Not checked	2024-01-20	2023-06-10	237	140
7	61	Not checked	2023-06-12	2023-11-05	279	195
8	73	Not checked	2023-12-02	2023-05-28	274	187
9	75	Not checked	2023-11-03	2023-12-03	300	130
10	83	Not checked	2023-10-04	2023-04-26	259	63
11	91	Not checked	2023-05-21	2023-09-03	234	76
12	92	Not checked	2024-01-26	2023-04-07	260	154
13	107	Not checked	2024-03-06	2023-05-01	218	188
14	110	Not checked	2023-07-16	2024-02-09	256	109
15	124	Not checked	2023-09-10	2024-01-09	226	24
16	152	Not checked	2023-08-23	2023-07-25	250	154
17	158	Not checked	2024-01-26	2023-07-05	245	33
18	200	Not checked	2024-01-22	2023-08-27	231	174
19	212	Not checked	2023-06-06	2023-12-25	246	7
20	213	Not checked	2023-08-07	2024-01-05	237	25
21	224	Not checked	2023-10-04	2023-09-22	263	123
22	239	Not checked	2023-10-23	2023-05-09	225	

Successfully run. Total query runtime: 40 msec. 56 rows affected

Total rows: 56 | Query complete 00:00:00.040

TASK 15

Object Explorer

Languages

Publications

Schemas

Subscriptions

lab4

Casts

Catalogs

Event Triggers

Extensions

Foreign Data Wrappers

Languages

Publications

Schemas (1)

public

Aggregates

Collations

Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (10)

airline

Columns (6)

airline_id integ

airline_code of

airline_name c

airline_countr

created_at dat

update_at date

lab4/postgres@Po...

lab4/postgres@PostgreSQL 18*

L2/postgres@Post...

lab4/postgres@PostgreSQL 18

Query

Query History

--task 15
SELECT * FROM baggage_check WHERE EXTRACT(YEAR FROM update_at)=EXTRACT(YEAR FROM created_at)
AND EXTRACT(MONTH FROM update_at) = EXTRACT(MONTH FROM created_at)
AND update_at < created_at;

Data Output

Messages

Notifications

Showing rows: 1 to 18

Page No: 1 of 1

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	34	Checked	2023-04-25	2023-04-08	28	57
2	86	Checked	2023-12-25	2023-12-07	312	22
3	104	Checked	2024-03-17	2024-03-14	41	83
4	109	Not checked	2023-08-30	2023-08-09	469	91
5	119	Not checked	2023-12-02	2023-12-01	135	55
6	148	Checked	2024-03-17	2024-03-16	86	5
7	219	Checked	2023-04-17	2023-04-14	233	200
8	248	Not checked	2023-06-22	2023-06-09	449	60
9	251	Not checked	2023-06-19	2023-06-16	260	26
10	299	Checked	2023-04-26	2023-04-08	385	191
11	304	Not checked	2023-07-29	2023-07-22	210	10
12	396	Not checked	2023-07-13	2023-07-01	37	144
13	412	Not checked	2023-09-30	2023-09-27	347	70
14	423	Not checked	2024-02-22	2024-02-10	449	148
15	444	Not checked	2023-05-18	2023-05-02	282	59
16	455	Not checked	2023-12-31	2023-12-05	321	159
17	474	Not checked	2024-01-28	2024-01-19	75	64
18	494	Checked	2023-06-13	2023-06-09	477	18

Successfully run. Total query runtime: 51 msec. 18 rows affected

Total rows: 18 Query complete 00:00:00.051 LF Ln 37, Col 28